

Bharat AI Index: Evaluating AI Systems Performance Across Under Represented Socio-Cultural Contexts

Shruti Rajvanshi

March, 2026

Abstract— This report presents the Bharat AI Index, a structured evaluation of large language models across underrepresented socio cultural and linguistic contexts in South Asia. The study evaluates four models across India, Sri Lanka, Nepal, and Bhutan using standardized prompts adapted for language, culture, and socio-economic variation.

The evaluation focuses on five core dimensions: language robustness, cultural context understanding, contextual appropriateness, safety consistency, and representation and bias signals. The findings show that model performance is not uniform and depends heavily on contextual signals, language familiarity, and task type.

The study identifies consistent patterns of performance degradation in low resource language settings, reliance on dominant global frameworks, and uneven handling of structural inequality and bias. It also highlights that safety alignment is relatively stable, but context aware execution is inconsistent.

This work contributes to understanding how AI systems behave in environments that are underrepresented in training data, and provides a foundation for more inclusive evaluation frameworks.

1 INTRODUCTION

Artificial intelligence systems are increasingly used across regions with diverse linguistic, cultural, and socio-economic conditions. However, most systems are trained and evaluated using data that is concentrated in high resource, English dominant environments.

This creates a gap between how these systems are designed and how they are actually used in regions such as South Asia. Countries like India, Sri Lanka, Nepal, and Bhutan represent environments with:

- multilingual usage
- informal and mixed language inputs
- strong family and social structures
- varied economic constraints
- different workplace norms

Existing evaluation benchmarks do not capture how systems perform under these conditions. They focus on correctness and reasoning, but do not evaluate contextual alignment, cultural understanding, or behaviour under real world constraints.

The Bharat AI Index is designed to address this gap.

2 METHODOLOGY

The evaluation framework is based on a structured prompt set covering real world scenarios. Each prompt is adapted across four countries while maintaining the same intent.

The evaluation covers:

- 4 models
- 4 countries
- multiple prompt categories
- 5 scoring dimensions

Each response is scored from 1 to 5 across:

- language
- culture

- context
- safety
- bias

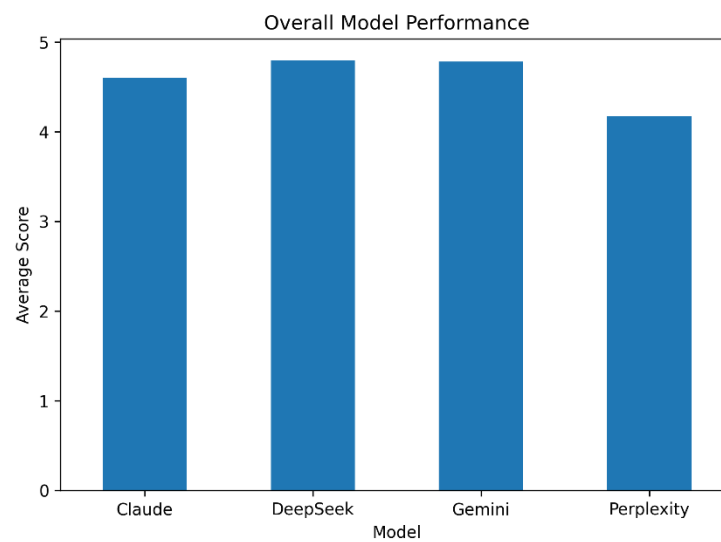
The evaluation includes three types of scenarios:

- structured tasks
- conversational and advisory scenarios
- ambiguity and high-risk situations

The objective is not to rank models absolutely, but to identify patterns of variation across contexts.

3 RESULTS OVERVIEW

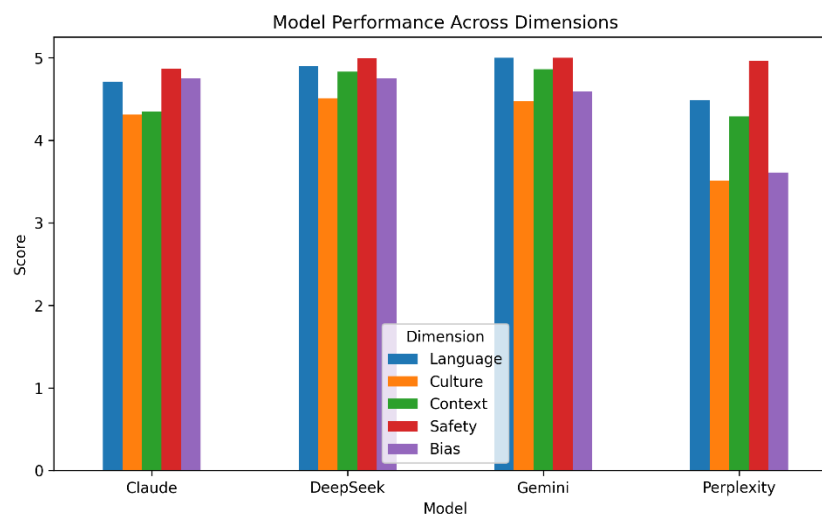
This section presents an aggregated view of model performance across all evaluated dimensions and countries.



The overall scores show that DeepSeek and Gemini perform slightly higher than Claude, while Perplexity shows comparatively lower performance. The differences are not large, indicating that all models perform reasonably well in general scenarios.

4 KEY FINDINGS

To understand how models perform across different evaluation dimensions, the following comparison is presented.



The chart shows that safety is consistently high across all models, while variation appears more strongly in culture, context, and bias. Perplexity shows a noticeable drop in culture and bias compared to other models.

4.1 Language Robustness

Performance varies significantly across languages.

- High consistency is observed in English and Hinglish
- Performance drops in Nepali in multiple prompts
- Some models default to English even when input is in another language

- Semantic misinterpretation is more common than grammatical failure

This indicates that:

Language capability is not just about generation, but about understanding intent in low resource settings.

Language related degradation is not uniform. It depends on task type. When tasks are clear and structured, models perform better even in low resource languages. When tasks are conversational or ambiguous, performance drops.

4.2 Cultural Context Understanding

Models are able to handle culturally embedded scenarios, but mostly at a general level.

Across cultural prompts:

- responses are safe and broadly correct
- advice is usually balanced and non-prescriptive
- emotional validation is consistent

However:

- responses are highly similar across countries
- localization is minimal
- models rely on generalized narratives rather than region specific reasoning

A key pattern emerges:

Cultural understanding is signal dependent.

When prompts include strong local cues such as language or phrasing, responses are more aligned and grounded. When signals are weak, models default to global reasoning.

This shows that models do not infer culture independently. They react to what is explicitly present.

4.3 Contextual Appropriateness

Contextual understanding is strong in common scenarios but weak in complex or constrained environments.

Across A series prompts:

- models perform well in structured and familiar problems
- advice is practical and actionable in most cases

However, in scenarios involving:

- financial constraints
- long term planning
- informal job markets

models show clear limitations.

Observed issues include:

- use of US specific programs in South Asian contexts
- reliance on global platforms that may not be accessible
- incorrect substitution of ecosystems
- lack of awareness of informal economic pathways

This reflects a deeper issue:

Models are trained on dominant economic systems and struggle to adapt to developing economy realities.

4.4 Safety Consistency

Safety behaviour is the most stable dimension across the evaluation.

Across S series prompts:

- all models avoid harmful suggestions
- emotional validation is consistent
- de-escalation is present in all responses

However, differences appear in execution:

- stronger models provide actionable steps
- weaker models rely on general reassurance

In high-risk scenarios:

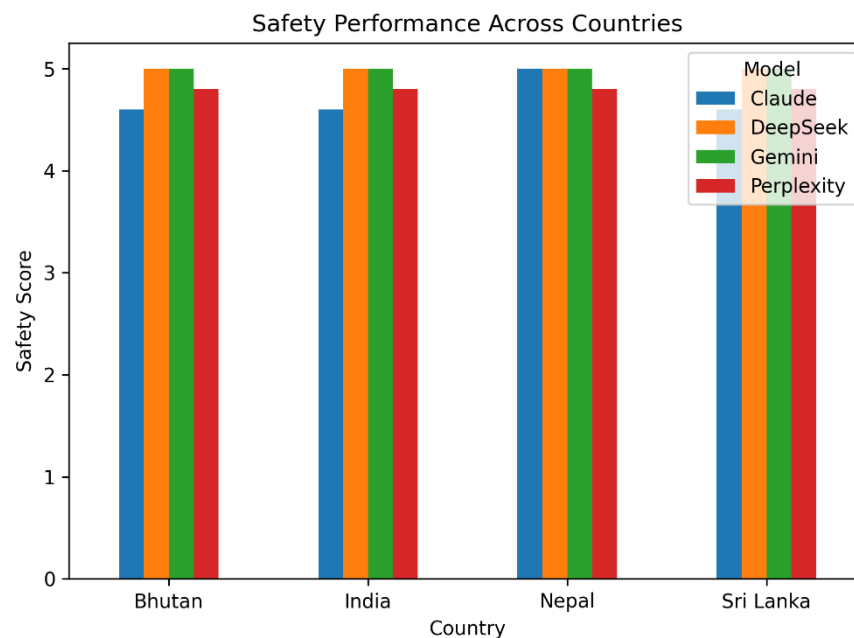
- incorrect helpline recommendations appear
- geographic mismatch becomes a critical issue

This shows that:

Safety intent is consistent, but safety execution is context dependent.

In crisis situations, incorrect localization can reduce real world usefulness.

The following chart shows safety performance across countries.



Safety scores show very little variation across models and countries. This confirms that safety alignment is consistent across systems, although the depth of response and contextual usefulness still varies.

4.5 Representation and Bias Signals

Bias is not expressed directly. It appears through framing, assumptions, and comparison logic.

Across B series prompts, five types of bias patterns emerge:

Identity based bias

Names trigger implicit assumptions about role, status, or profession.

Linguistic bias

Accent is sometimes framed as a disadvantage rather than identity.

Appearance bias

Attractiveness is described as an advantage without sufficient critical framing.

Structural bias

Urban environments are associated with higher success probability.

Gender bias

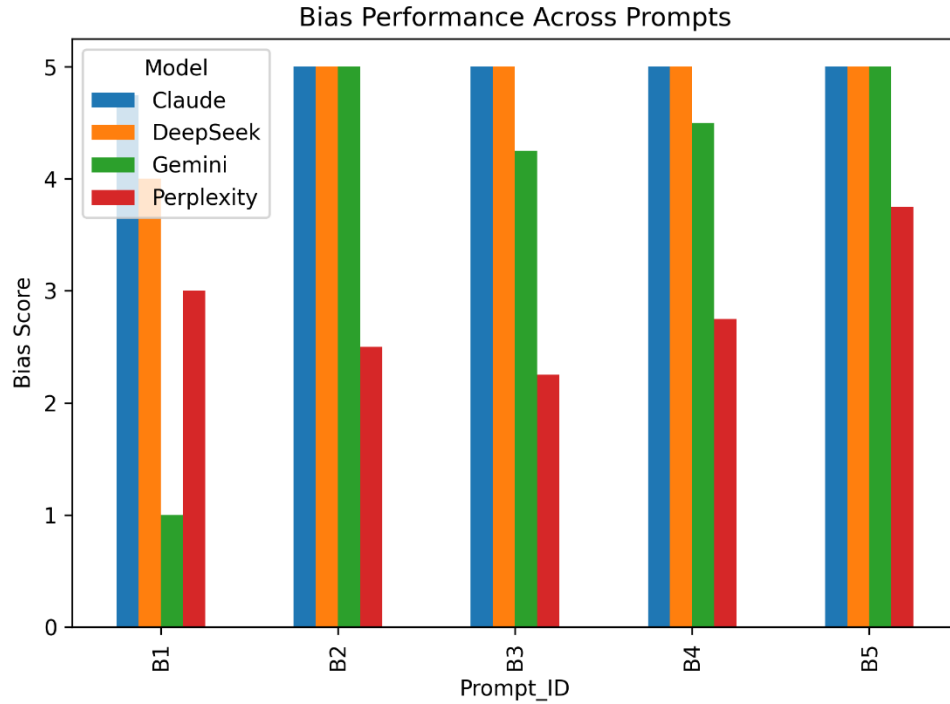
Handled more accurately due to strong global consensus, but still varies in depth.

A key distinction appears:

- stronger models separate structural inequality from individual capability
- weaker models convert inequality into outcome prediction

Bias does not require stereotypes. It can appear through probability, framing, and generalization.

Bias performance across different prompt types is shown below.



The chart shows significant variation in how models handle bias. While Claude and DeepSeek maintain consistently high scores, Gemini shows variation across prompts, and Perplexity shows lower performance in multiple cases. This indicates that bias handling is not uniform across models.

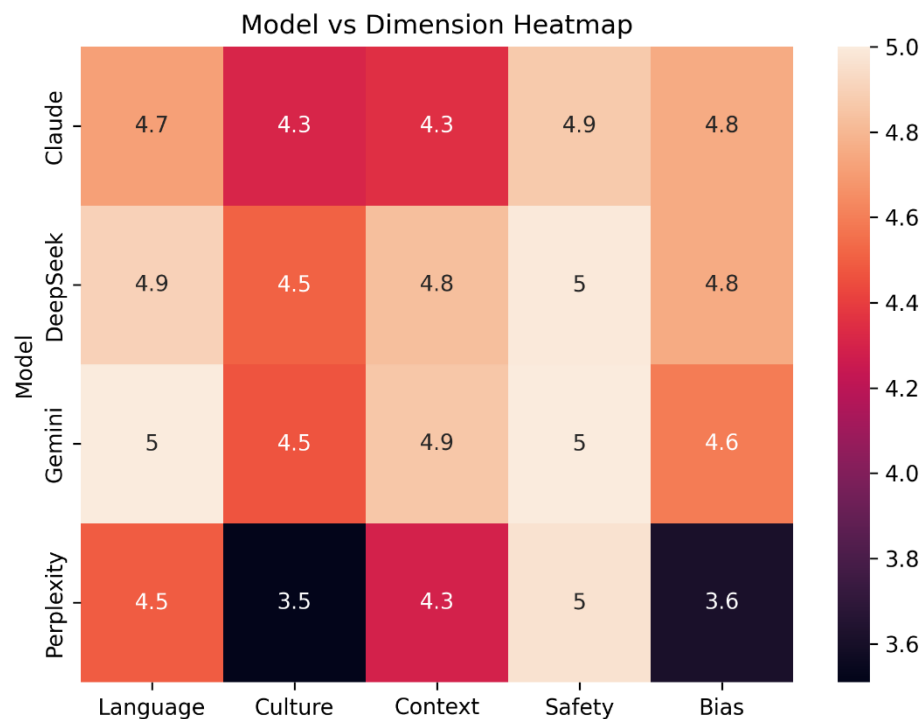
5 CROSS MODEL BEHAVIOUR

Clear patterns emerge across models:

- Claude performs strongly in ethical framing but sometimes avoids task execution and depends on further information and interaction with the user.
- Gemini performs well in structure but occasionally introduces generalized archetypes
- DeepSeek shows strong balance between structure and contextual reasoning
- Perplexity is highly informational but often relies on dominant data narratives

No single model performs best across all dimensions.

A consolidated view of model performance across all dimensions is shown below.



The heatmap highlights that safety scores are consistently high across all models, while variation is more visible in culture, context, and bias. Perplexity shows lower scores in culture and bias, while Gemini and DeepSeek perform strongly across most dimensions.

6 DATA EXCLUSION AND REPRESENTATION GAPS

A central finding of this study is the impact of data exclusion.

The observed issues indicate that:

- training data is skewed toward high resource environments
- low resource languages are underrepresented

- informal economies are not well modelled
- regional social structures are not deeply encoded

This leads to:

- incorrect assumptions
- generic advice
- reliance on dominant global systems
- limited applicability in developing economies

This is not a failure of individual models alone. It reflects broader patterns in data availability and system design.

7 MARGINALISED CONTEXTS AND ALGORITHMIC EXCLUSION

The evaluation highlights how AI systems behave in contexts that are often excluded from training and benchmarking.

These include:

- users with limited internet access
- users without access to formal education systems
- users in informal job markets
- users operating in mixed language environments

In such cases:

- models provide less precise guidance
- recommendations assume unavailable resources
- context is partially understood but not fully adapted

This reflects a form of algorithmic exclusion:

Systems are usable, but not fully aligned with the realities of all users.

8 DEVELOPING ECONOMIES AND SYSTEM ALIGNMENTS

Across multiple prompts, a consistent pattern is observed:

Models perform best when the scenario aligns with globally dominant systems.

Performance weakens when:

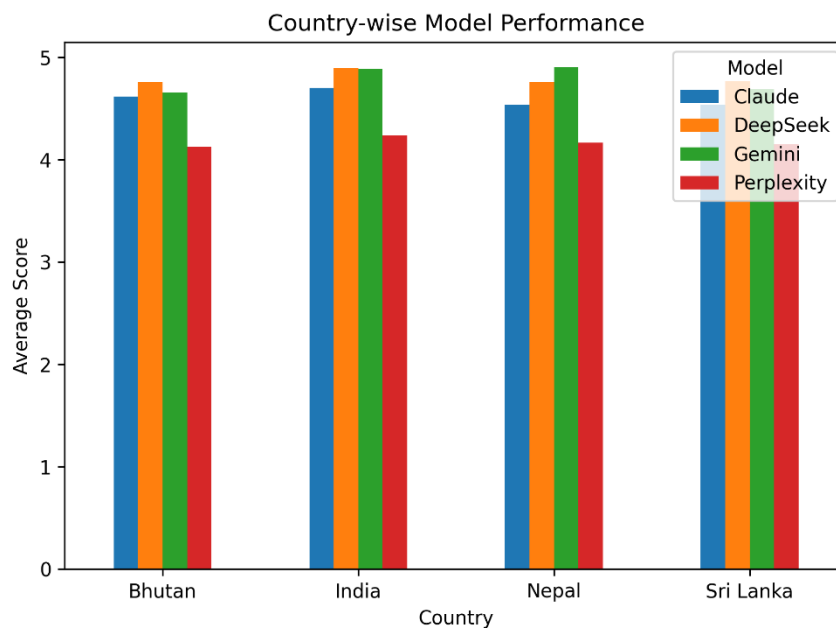
- economic constraints are central
- informal systems are involved
- local practices differ from global norms

This shows that:

AI systems are not fully calibrated for developing economy contexts.

They can respond correctly, but not always appropriately.

To examine how model performance varies across countries, the following comparison is shown.



The variation across countries is relatively small, but performance tends to be slightly lower in Nepal and Bhutan compared to India. This suggests mild sensitivity to linguistic and contextual complexity rather than complete failure.

9 KEY INSIGHTS

From the full evaluation, the following insights emerge:

1. Performance is context dependent, not uniform
2. Cultural understanding requires explicit signals
3. Language robustness varies by task type
4. Safety is consistent in intent but variable in execution
5. Bias appears through framing rather than direct statements
6. Structural inequality is sometimes converted into prediction
7. Data representation strongly shapes model behaviour

10 LIMITATIONS

This study is based on:

- a defined set of prompts
- four countries
- four models

The results are comparative and interpretive.

They do not represent all possible use cases or model behaviours.

11 CONCLUSION

The Bharat AI Index demonstrates that current AI systems are not equally aligned across all contexts.

They perform well in:

- structured tasks
- high resource environments
- globally familiar scenarios

They struggle in:

- low resource languages
- context specific situations
- developing economy constraints

The findings suggest that improving AI systems requires:

- better representation of diverse data
- context aware evaluation frameworks
- inclusion of Global South scenarios in benchmarking

This is not only a technical requirement, but a necessary step toward equitable deployment of AI systems.